

Vol 1 Chapter 6 — The Operator Zoo, In Detail

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Chapter 6 — The Operator Zoo, in Detail

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Volume 0 Chapter 7 introduced the substrate's operator zoo — about a dozen substrate-native operators on $H^2(D_{IV}^5)$, each derived from a specific feature of the substrate's symmetry structure. This chapter is the operational reference: each operator written out with its explicit Bergman-side realization, its commutators with the others, its eigenvalue spectrum on the K-type lattice. After this chapter we have the apparatus to write dynamics in Chapter 7.

6.1 The common Hilbert space

Every operator acts on $H^2(D_{IV}^5)$ (Chapter 2). The K-type decomposition under $K = SO(5) \times SO(2)$ provides eigenvalues:

$$C_2(\lambda) = \langle \lambda + \rho, \lambda + \rho \rangle - \langle \rho, \rho \rangle, \quad \rho = (5/2, 3/2).$$

The lowest non-trivial K-type $V_{(1,1)}$ has $C_2 = 6$ (BST primary). All higher K-types carry BST-primary-derivable rational eigenvalues.

6.2 Position and momentum

Position acts by multiplication, momentum by Wirtinger derivative:

$$(\hat{X}f)(z) = z \cdot f(z), \quad (\hat{P}f)(z) = -i \partial_z f(z).$$

Commutator from Bergman reproducing structure (Chapter 2):

$$[\hat{X}, \hat{P}] = i\hbar.$$

Position spectrum aligns with the perfect numbers $\{6, 28, 496, 8128, \dots\}$ — Elie’s K71 result (T2419).

6.3 Angular momentum, spin, total

From $SO(5)$ generators and the $\text{Pin}(2)$ intrinsic representation:

$$[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k, \quad [\hat{S}_i, \hat{S}_j] = i\hbar\epsilon_{ijk}\hat{S}_k, \quad [\hat{L}_i, \hat{S}_j] = 0.$$

Eigenvalues: $j(j+1)\hbar^2$ for $\hat{L}^2$ and $s(s+1)\hbar^2$ for $\hat{S}^2$. Three-dimensional laboratory angular momentum is the $SO(3) \subset SO(5)$ restriction.

6.4 The Hamiltonian

$\hat{H}$ is the Casimir of $SO_0(5, 2)$ on $L^2(D_{IV}^5; L_\lambda)$. Lowest non-trivial eigenvalue $C_2 = 6$ (Elie K52a Session 29). The substrate’s natural energy unit is set here; particle masses in Volume 2 are dimensionless multiples.

6.5 Charge and chirality

$\hat{Q}$ is $SO(2)$ weight on scalar K-types (Lyra T2470). Integers on lepton/boson K-types; $\pm 1/N_c$ fractions on quark K-types via N_c -fold color sub-structure.

$\hat{\gamma}^5$ is $SO(2)$ half-weight phase on spinor K-types (Lyra T2471). Eigenvalues ± 1 , involutive.

6.6 Discrete-symmetry operators

$\hat{P}$ (Möbius lift, T2472): unitary involution. $\hat{P}\hat{X}\hat{P}^{-1} = -\hat{X}$, $\hat{P}\hat{P}\hat{P}^{-1} = -\hat{P}$ (momentum), $\hat{P}\hat{L}\hat{P}^{-1} = +\hat{L}$.

$\hat{T}$ (Klein anti-unitary, T2433): $(\hat{T}f)(z) = \overline{f(\bar{z})}$. Kramers $\hat{T}^2 = \pm 1$ via $\text{Pin}(2)$.

$\hat{C}$ ($SO(2)$ weight negation, T2434): unitary, $\hat{C}\hat{Q}\hat{C}^{-1} = -\hat{Q}$.

Composite $\hat{C}\hat{P}\hat{T}$ commutes with substrate Hamiltonian universally.

6.7 Bell–CHSH and number

$\hat{B}$ on bipartite states: $\text{Tr}(\hat{B}^2) = 126/16 = 7.875$, with sub-Tsirelson deviation $1/8 = 1/2^{N_c}$ — substrate signature falsifier. [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): $S = 2.82759$

± 0.00051 , 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]

$\hat{N}$ counts substrate cycles. Particles are substrate cycles (T1922).

6.8 The zoo at a glance

Op	Source	Commutates with $\hat{H}$?	Spectrum
$\hat{X}$	coset $\mathfrak{m}$	no	perfect numbers cluster
$\hat{P}$	coset $\mathfrak{m}$	yes if free	Wirtinger continuous
$\hat{L}, \hat{S}$	$SO(5)$, $\text{Pin}(2)$	yes (rotational)	$j(j+1)\hbar^2$
$\hat{H}$	$SO_0(5, 2)$ Casimir	self	$C_2 = 6$ min
$\hat{Q}$	$SO(2)$ weight	yes	$\mathbb{Z}, \pm 1/N_c$
$\hat{\gamma}^5$	$SO(2)$ half-weight	massless: yes	± 1
$\hat{P}, \hat{T}, \hat{C}$	discrete involutions	strong+EM yes; weak no	± 1
$\hat{C}\hat{P}\hat{T}$	composite	universal	± 1
$\hat{B}$	cycle correlator	invariant	$\text{Tr}(\hat{B}^2) = 126/16$ [FIRED — bracket 10:09 2026-09-13 (Grace, R143 G2; register E4, certified K1893): the sub-Tsirelson ceiling / SCMP Layer-1 prediction quoted here was refuted by Poh et al., PRL 115, 180408 (2015): S $= 2.82759 \pm$ 0.00051 , 41.9σ above BST's 2.806. This sentence is a May-2026 record, not a standing claim; nothing else on the line is changed.]
$\hat{N}$	cycle counter	conserved	$\mathbb{Z}_{\geq 0}$

6.9 What comes next

Chapter 7 uses the zoo to develop substrate dynamics.

Where to look this up: Per-operator T-numbers as cited. For standard-QFT operator algebra, Peskin and Schroeder Chapters 2–4. Wigner anti-unitary theorem in Streater–Wightman, Chapters 1–2.